

AWS Elastic Beanstalk Kickstart

Introduction

AWS Elastic Beanstalk is a **Platform as a Service (PaaS)** offered by Amazon Web Services that simplifies the deployment and management of web applications. Instead of manually configuring servers, installing software, and managing infrastructure, developers can upload their application, and Elastic Beanstalk automatically provisions the required AWS resources.

For this project, I deployed the **Sample Application** provided by AWS using a **Single Instance** environment.

Project Objective

The objective of this project was to understand how AWS Elastic Beanstalk simplifies application deployment by automatically creating and configuring the required cloud resources.

This deployment helps beginners understand the basic workflow of hosting an application on AWS without manually configuring an EC2 instance.

AWS Services Used

1. AWS Elastic Beanstalk

Elastic Beanstalk is responsible for managing the complete deployment process.

It automatically:

- Creates the application environment
- Launches an EC2 instance
- Deploys the application
- Monitors the application health
- Provides an Environment URL for accessing the application

In this project, Elastic Beanstalk handled all infrastructure management automatically.

2. Amazon EC2

Elastic Beanstalk launched **one Amazon EC2 instance** because the environment type selected was **Single Instance**.

The EC2 instance is the virtual server where the sample application is hosted.

Instead of manually creating and configuring the server, Elastic Beanstalk performed these tasks automatically.

3. IAM (Identity and Access Management)

IAM provides secure access between AWS services.

Elastic Beanstalk uses IAM Roles to:

- Launch EC2 instances
- Access AWS resources
- Perform deployment operations securely

No manual IAM configuration was required because the default Elastic Beanstalk roles were used.

4. Amazon CloudWatch

CloudWatch continuously monitors the health of the Elastic Beanstalk environment.

It provides:

- Environment Health Status
- Instance Monitoring
- Deployment Events
- System Metrics

This helps identify whether the application is running successfully.

Environment Type

The application was deployed using a **Single Instance Environment**.

Characteristics

- One EC2 Instance
- No Load Balancer
- Auto Scaling Group
- Suitable for learning, development, and testing
- Lower cost compared to production environments

Deployment Workflow

Developer

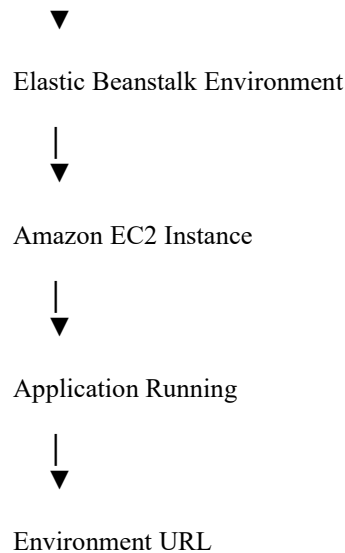


Elastic Beanstalk Application



Sample Application





Benefits of Using Elastic Beanstalk

- Easy application deployment
- No manual server configuration
- Automatic EC2 provisioning
- Built-in environment monitoring
- Simplified application management
- Beginner-friendly cloud deployment platform

Learning Outcomes

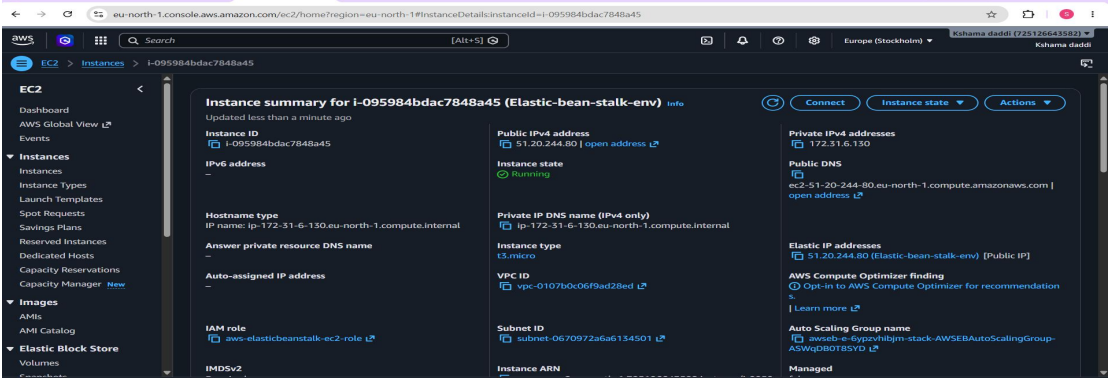
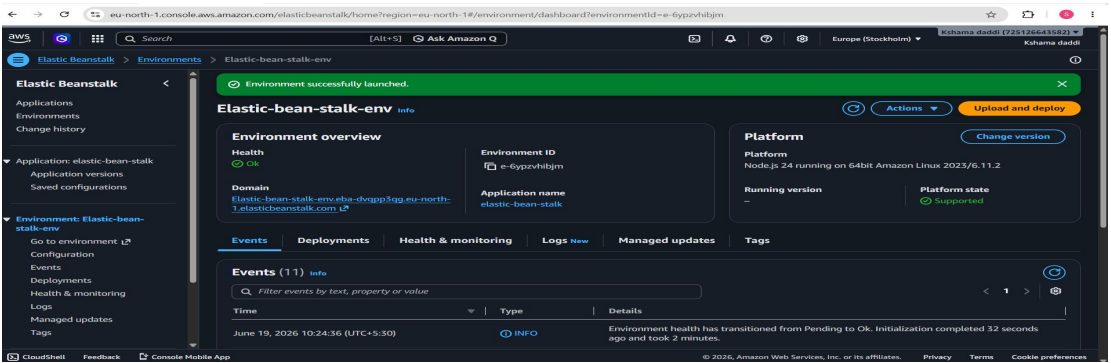
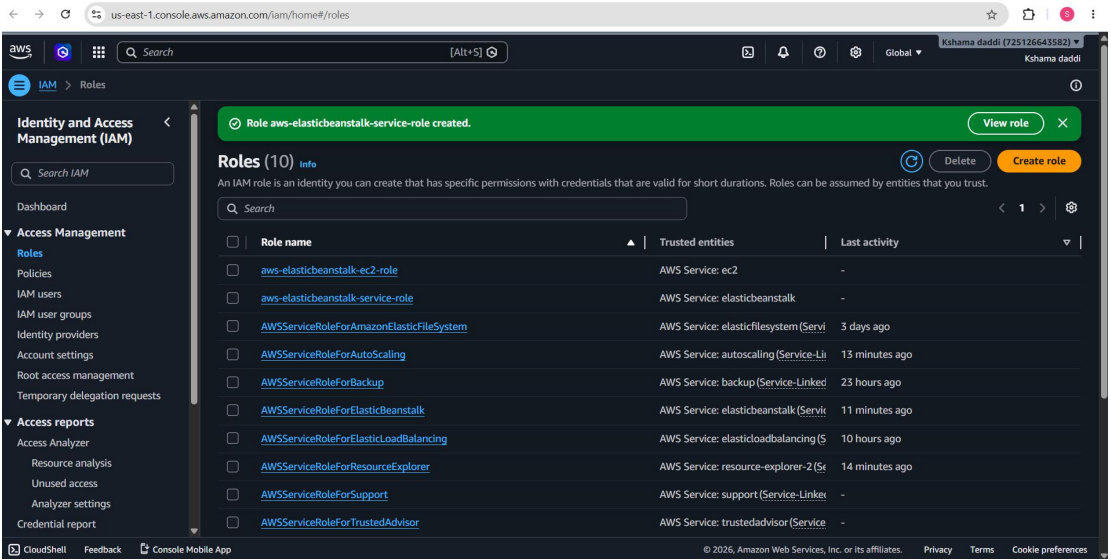
After completing this project, I learned how to:

- Create an Elastic Beanstalk Application
- Deploy the AWS Sample Application
- Understand the purpose of a Single Instance Environment
- Observe how Elastic Beanstalk automatically creates an EC2 instance
- Access the application through the Environment URL
- Monitor the deployment using the Elastic Beanstalk Dashboard
- Understand the basic architecture of AWS Elastic Beanstalk

Conclusion

This project provided a practical introduction to cloud application deployment using AWS Elastic Beanstalk. By deploying the Sample Application in a Single Instance environment, I gained an understanding of how Elastic Beanstalk automates infrastructure provisioning and application deployment while reducing the complexity of managing cloud resources. It serves as an excellent starting point for learning AWS application hosting before moving on to more advanced deployments involving Load Balancers, Auto Scaling, custom domains, and CI/CD pipelines.

Screenshoots:



eu-north-1.console.aws.amazon.com/ec2/home?region=eu-north-1#AutoScalingGroupsId=awseb-e-6ypzvhibjm-stack-AWSEBAutoScalingGroup-ASWqDB0T8SYD:view=details

Auto Scaling groups (1/1) info

Search your Auto Scaling groups

Name	Status	Launch template/configuration	Instances	Instance health	Desired capacity
awseb-e-6ypzvhibjm-stack-AWSEBAutoScalingGroup-ASWqDB0T8SYD	At desired capacity	AWSEBC2LaunchTemplate-jtkOURmJnx	1	1/1 Healthy	1

Auto Scaling group: awseb-e-6ypzvhibjm-stack-AWSEBAutoScalingGroup-ASWqDB0T8SYD

Details Integrations Automatic scaling Instance management Instance refresh Activity Monitoring Tags

awseb-e-6ypzvhibjm-stack-AWSEBAutoScalingGroup-ASWqDB0T8SYD Capacity overview

arn:aws:autoscaling:eu-north-1:725126643582:autoScalingGroup:98a85d1b-9956-4727-935d-29510f4df3e:autoScalingGroupName/awseb-e-6ypzvhibjm-stack-AWSEBAutoScalingGroup-ASWqDB0T8SYD

Desired capacity	Scaling limits	Desired capacity type	Status
Units (number of instances)	Units (number of instances)	Units (number of instances)	Units (number of instances)
1	1 - 1	At desired capacity	At desired capacity

elastic-bean-stalk-env.eba-dvqpp3gg.eu-north-1.elasticbeanstalk.com

AWS Elastic Beanstalk

Welcome to Your Elastic Beanstalk Application

Congratulations! Your Node.js application is now running on your own dedicated environment in the AWS Cloud.

Learn More

Benefits of AWS Elastic Beanstalk

Discover why thousands of developers rely on AWS Elastic Beanstalk to deploy and manage their applications.

